WIRELESS EXTENDER FOR IR REMOTE CONTROLS

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Using this system, you can control your IR-remote-controlled home appliances like DVD player and air-conditioner from another room without the hassle of running extra cables between the two rooms. The system wirelessly sends infrared signals from one room to another.

The system is a combination of an infrared transmitter unit and a receiver unit, both housed in separate small plastic boxes. Keep the transmitter unit in the room where you are sitting with the remote control of the appliance to be controlled (say, a DVD player), while the receiver unit has to be kept on or nearby the DVD player in another room. The transmitter detects the IR pulses from your remote and retransmits on the UHF band at 433 MHz.

The receiver unit (in another room) picks up the transmitted signals, detects and demodulates them with the 38kHz ultrasonic carrier used by IR remote systems. IR pulses from a pair of IR LEDs (IR LED1 and IR LED2) provide controlling signals to the DVD player located in the other room.

Fig. 1 shows the transmitter circuit. The codes from the remote control are picked up by the IR receiver module (TSOP1738). The detected control codes pass through gates N4 and N1 of IC CD4093 (IC1) to the data input of UHF transmitter module (TX1), which transmits them to the receiver. Gate N2 drives LED1 to give a visual indication when TSOP1738 receives pulses from the IR remote control.

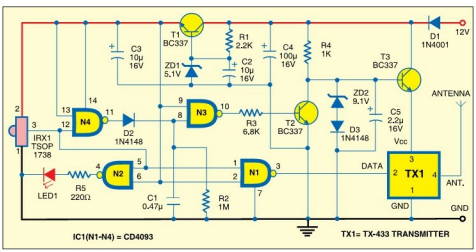


Fig. 1: Transmitter circuit

The combination of zener diode ZD1 and transistor T1 (with R1, C2 and C3) provides regulated 5V DC from the unregulated 12V input. This 5V supply is used for the IR receiver module and quad Schmitt gate CD4093 (IC1). Diode D1 provides reverse-polarity protection, while capacitor C4 provides additional smoothing and acts as a reservoir.

The regulated 9V transmitter module (TX1) supply is arranged such that it enables the transmitter module only when IR remote code pulses need to be transmitted. It is achieved through the combination of transistor T3 and zener diode ZD2 with its series diode D3.

Fig. 2 shows the receiver circuit. A regulator built around transistor T4 and zener diode ZD3 provides regulated 5V DC to the receiver module (RX1) and CD4093 quad Schmitt NAND gate (IC2).

When the receiver module detects a signal from the transmitter, it outputs the code pulses to pin 1 of gate 1. The 38kHz pulses generated by the

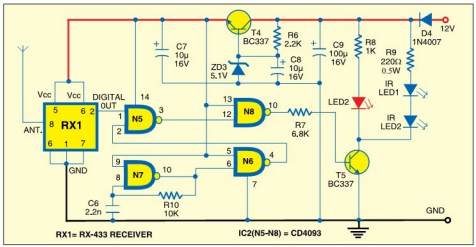


Fig. 2: Receiver circuit